B.Tech. Degree II Semester Regular Examination July 2024

CE/ME/SE 23-200-0202A ENGINEERING PHYSICS

(2023 Scheme)

Time: 3 Hours | Maximum Marks: 50

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Interpret modern devices and technologies based on lasers and optical fibres.

CO2: Explain the basic principles of crystal physics.

CO3: Summarise the characteristics and applications of superconducting materials, nano materials and smart materials.

CO4: Illustrate the theory of semiconductors and magnetic materials.

CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(5 \times 2 = 10)$	Marks	BL	CO	PO
I.	(a)	Draw the energy level diagram for the He-Ne laser and mark all the transitions.	2	L1	1	1
	(b)	Obtain a relation between density and lattice constant for a material having the simple cubic structure.	2	L2	2	1
	(c)	State any two applications of metallic glasses.	2	L3	3	1
	(d)	Briefly elaborate the term surface occupancy.	2	L2	3	1
	(e)	Bring out the relevance of majority and minority carriers in semiconductors.	2	L2	4	2

PART B

		$(4 \times 10 = 40)$				
II.	(a)	Describe the apparatus, principle and working of a three level laser system with suitable diagrams	7	L2	1	1
	(b)	All excited states are metastable states. True or false. Support your answer.	3	L4	1	2
		OR				
III.	(a)	Explain the difference in the propagation of a signal in a step index and graded index optical fibre with suitable figures.	7	L2	1	1
	(b)	The numerical aperture of an optical fibre is found to be 0.2441 and the refractive index of the core is 1.5. Calculate the refractive index of the cladding and the acceptance angle.	3	L3	1	2

(P.T.O.)

			Marks	BL	CO	PO
IV.	(a)	Define the term Miller indices. How do you find the Miller indices of a plane parallel to the x and z axis and having intercepts of 1/3 along the y axis for a cubic system? State any two characteristics of Miller indices.	7	L2	2	1
	(b)	Draw the (020), (101) and (242) planes for an unit cell of the simple cubic structure.	3	L2	2	2
		OR				
V.	(a)	Describe with the help of diagrams how X-rays are produced.	7	L1	2	1
	(b)	Find the angles at which 1.44 $\mathring{A}$ x-rays are diffracted by a polished crystal of rock salt (d = 2.8 $\mathring{A}$), forming the first order and second order maxima.	3	L3	2	2
VI.	(a)	Explain the working of a shape memory alloy with diagrams and give two of its applications.	7	L1	3	1
	(b)	Lead has a transition temperature of 8.25 K. If its characteristic critical field at 0 K is 12.5×10^5 A/m, calculate the critical current density of a lead wire of diameter 2 mm at 5 K.	3	L3	3	1
		OR				
VII.	(a)	Define the terms (i) Isotope effect. (ii) Messier effect. (iii) Cooper pair. Wherever possible support with diagrams.	7	L1	3	1
	(b)	Elaborate the term fullerenes with examples.	3	L2	3	2
VIII.	(a)	Explain in detail how magnetic materials are classified with examples.	7	L1	4	1
	(b)	The magnetic field applied in a diamagnetic material is 10000 A/m. Calculate the magnetisation and flux density of the material which is having a susceptibility of -0.4×10^{-5}.	3	L3	4	2
		OR				
IX.	(a)	Using the band gap theory differentiate between the three groups of materials based on conductivity.	7	L1	4	1
	(b)	Carbon, silicon and germanium have the same lattice structure, however only carbon is not a semiconductor. Why?	3	L4	4	2

Blooms's Taxonomy Levels
L1 – 29%, L2 – 38%, L3 – 24%, L4 – 9%.

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